

RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)

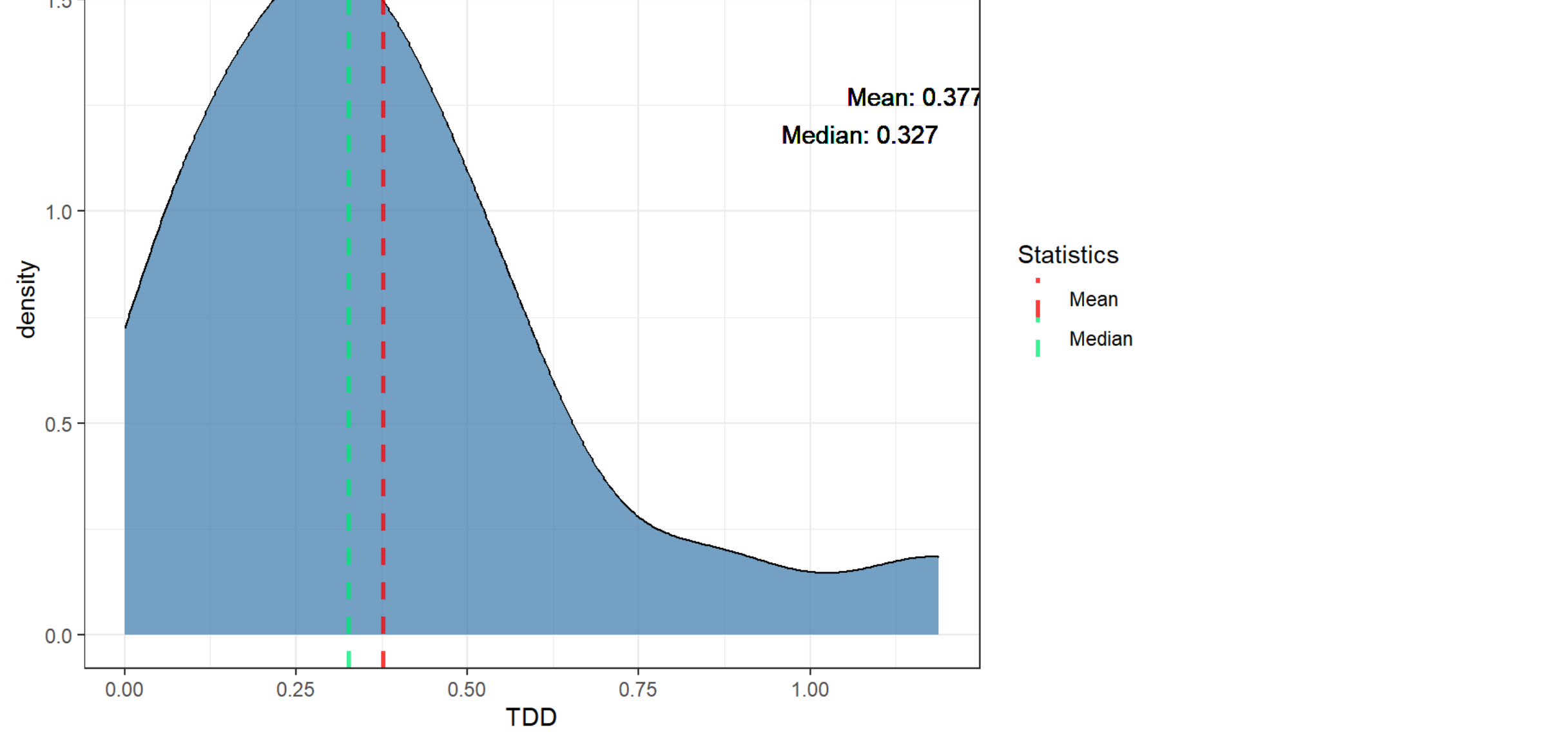
## # Rows: 1,641
## # Columns: 11
## $ repo                <chr> "commons-collections", "commons-collections", "com...
## $ pr_number            <int> 198, 203, 203, 203, 203, 203, 203, 203, ...
## $ rule                 <chr> "java:S125", "java:S2975", "java:S112", "java:S116...
## $ severity             <chr> "MAJOR", "BLOCKER", "MAJOR", "MAJOR", "MA...
## $ file                 <chr> "src/test/java/org/apache/commons/collections4/map...
## $ type                 <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SML...
## $ status               <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...
## $ debt                 <int> 5, 30, 20, 30, 20, 5, 0, 20, 5, 0, 20, 5, 0, 20, 2...
## $ ncloc_affected_file  <int> 45, 189, 189, 189, 189, 189, 189, 189, 1...
## $ complexity           <int> 8, 38, 38, 38, 38, 38, 38, 38, 38, 38, 38, 38, ...
## $ origin               <chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 18 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 18 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

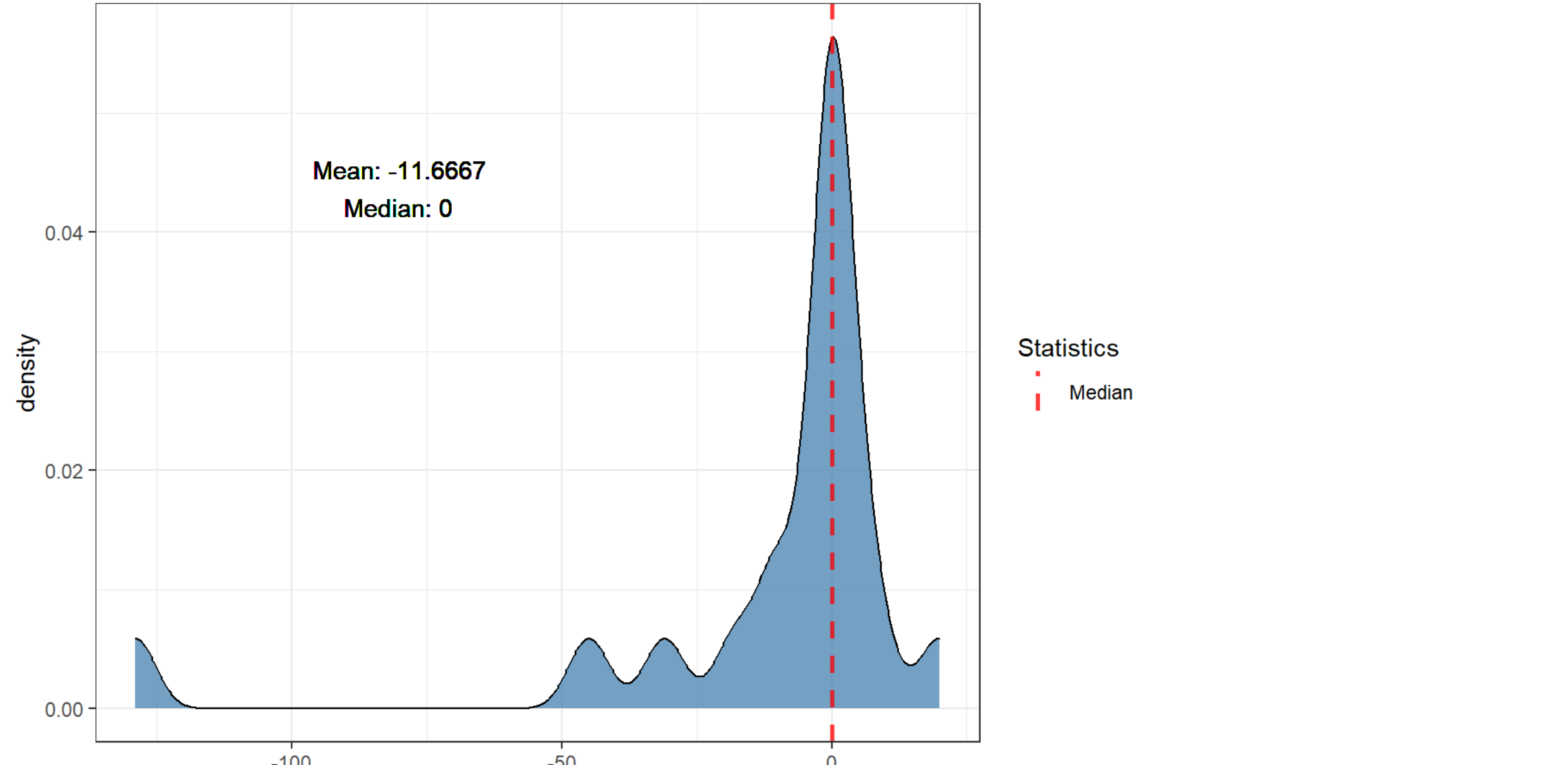


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 18 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 18 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 x 4
##   repo                tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 commons-collectio...      33.3             50             16.7
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 x 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##   <dbl>          <dbl>          <dbl>
## 1      33.3             50             16.7
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 x 3
##   repo                preexisting_tdv_equal_to_zero preexisting_tdv_greater_th...
##   <chr>                <dbl>          <dbl>
## 1 commons-collections      50             50
## # i abbreviated name: 'preexisting_tdv_greater_than_zero'
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##   <dbl>          <dbl>
## 1      50             50
```

Percentages for new TDV by repo

```
## # A tibble: 1 x 3
## # Groups:   repo [1]
##   repo                new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>
## 1 commons-collections      80             20
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##   <dbl>          <dbl>
## 1      80             20
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_j$$

given a rule *i*, its *PositionScore_i* will be the sum of its position across all *k* repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 x 2
## # Groups:   rank [10]
##   rank `commons-collections`
##   <int> <chr>
## 1     1 java:S112 - 10
## 2     2 java:S1170 - 6
## 3     3 java:S1874 - 5
## 4     4 java:S3008 - 5
## 5     5 java:S125 - 4
## 6     6 java:S1134 - 4
## 7     7 java:S1192 - 3
## 8     8 java:S1126 - 2
## 9     9 java:S3011 - 1
## 10    10 java:S1905 - 1
```

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

```
## # A tibble: 10 x 2
##   rule                position_score
##   <chr>          <dbl>
## 1 java:S112          122
## 2 java:S1170         123
## 3 java:S1874         124
## 4 java:S3008         125
## 5 java:S1134         126
## 6 java:S125          127
## 7 java:S1192         128
## 8 java:S1126         129
## 9 java:S1172         130
## 10 java:S1905        131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 x 2
## # Groups:   rank [10]
##   rank `commons-collections`
##   <int> <chr>
## 1     1 java:S1192 - 295
## 2     2 java:S125 - 207
## 3     3 java:S108 - 173
## 4     4 java:S1117 - 109
## 5     5 java:S2293 - 78
## 6     6 java:S2201 - 46
## 7     7 java:S3878 - 45
## 8     8 java:S1854 - 36
## 9     9 java:S112 - 35
## 10    10 java:S1133 - 33
```

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

```
## # A tibble: 10 x 2
##   rule                position_score
##   <chr>          <dbl>
## 1 java:S1192          122
## 2 java:S125           123
## 3 java:S108           124
## 4 java:S1117          125
## 5 java:S2293          126
## 6 java:S2201          127
## 7 java:S3878          128
## 8 java:S1854          129
## 9 java:S112           130
## 10 java:S1123         131
```

Outliers

Higher TDD

```
## # A tibble: 18 x 5
##   repo                pr_number total_debt total_ncloc   tdd
##   <chr>          <int>    <int>    <int>    <dbl>
## 1 commons-collections    304      120      101  1.19
## 2 commons-collections    300      150      175  0.857
## 3 commons-collections    215       25       45  0.556
## 4 commons-collections    317       32       58  0.552
## 5 commons-collections    361      364      673  0.541
## 6 commons-collections    371     3229     7736  0.417
## 7 commons-collections     69      199      491  0.405
## 8 commons-collections    360      651     1894  0.344
## 9 commons-collections    280     1185     3610  0.328
## 10 commons-collections    203     1571     4822  0.326
## 11 commons-collections    305       85      283  0.300
## 12 commons-collections    295     6357    24020  0.265
## 13 commons-collections    256       8       34  0.235
## 14 commons-collections    315     109     665  0.164
## 15 commons-collections    198       5      45  0.111
## 16 commons-collections    398      33     324  0.102
## 17 commons-collections    400      18     290  0.0621
## 18 commons-collections    329       6      173  0.0347
```

Higher TDV

```
## # A tibble: 18 x 3
##   pr_number repo                tdv
##   <int> <chr>          <int>
## 1     215 commons-collections      0
## 2     256 commons-collections      8
## 3     295 commons-collections      5
## 4      69 commons-collections      0
## 5     198 commons-collections      0
## 6     280 commons-collections      0
## 7     300 commons-collections      0
## 8     304 commons-collections      0
## 9     329 commons-collections      0
## 10    360 commons-collections      0
## 11    398 commons-collections      0
## 12    400 commons-collections     -10
## 13    203 commons-collections     -10
## 14    361 commons-collections     -18
## 15    315 commons-collections     -18
## 16    317 commons-collections     -31
## 17    305 commons-collections     -45
## 18    371 commons-collections    -129
```

Lower TDV

```
## # A tibble: 18 x 3
##   pr_number repo                tdv
##   <int> <chr>          <int>
## 1     371 commons-collections    -129
## 2     305 commons-collections     -45
## 3     317 commons-collections     -31
## 4     315 commons-collections     -18
## 5     203 commons-collections     -10
## 6     361 commons-collections     -10
## 7      69 commons-collections      0
## 8     198 commons-collections      0
## 9     280 commons-collections      0
## 10    300 commons-collections      0
## 11    304 commons-collections      0
## 12    329 commons-collections      0
## 13    360 commons-collections      0
## 14    398 commons-collections      0
## 15    400 commons-collections      0
## 16    295 commons-collections      5
## 17    256 commons-collections      8
## 18    215 commons-collections     20
```

Extra

Number of PRs with pre-existing TD

```
##   n
## 1 16
```

Number of java:S1192 issues that affect test code

```
##   n
## 1 298
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 0
```

Summary NCLOC

```
##   ncloc
## Min.   : 34
## 1st Qu.: 119
## Median : 307
## Mean   : 2318
## 3rd Qu.: 1589
## Max.   : 28812
```